As we learned in the previous chapter, the probability function in the second table is a discrete probability function because it has a finite domain—the integers from 1 to 10, inclusive. A discrete probability function has a finite domain or an infinite domain that can be listed. For example, if we flip a coin until we get heads, and let the random variables be the number of flips, then the domain is 1, 2, 3, 4, ... On the other hand, the distribution of heights (in inches) of college women includes infinitely many possible measurements, such as 53, 54.2, 66.5, $72.\overline{3}$, and so on, *within some real number interval.* Probability functions with such domains are called *continuous probability distributions.*

Continuous Probability Distribution

A **continuous random variable** can take on any value in some interval of real numbers. The distribution of this random variable is called a **continuous probability distribution**.

FOR REVIEW

The connection between area and the definite integral is discussed in the chapter on Integration. For example, in that chapter we solved such problems as the following.

Find the area between the x-axis and the graph of $f(x) = x^2$ from $x = 1$ to $x = 4$.

Answer: $\int_1^4 x^2 dx = 21$

Some probability functions are inherently discrete. For example, the number of owls in a forest preserve must be an integer, such as 0, 1, or 2, and could never take on any value in between. But the bird feeder example discussed earlier is different, because you could think of it as a simplification of a continuous distribution. It would have been possible to time the feedings with greater precision—to the nearest tenth of a minute or even to the nearest second (1/60 of a minute), if desired. Theoretically, at least, t could take on any positive real-number value between, say, 0 and 11 minutes. The graph of the probabilities $f(t)$ of these feeding times can be thought of as the continuous curve shown in Figure 1. As indicated in Figure 1, the curve was derived from our table by connecting the points at the tops of the bars in the corresponding histogram and smoothing the resulting polygon into a curve.

For a discrete probability function, the area of each bar (or rectangle) gives the probability of a particular feeding time. Thus, by considering the possible feeding times t as all the real numbers between 0 and 11, the area under the curve of Figure 2 between any two values of t can be interpreted as the probability that a feeding time will be between those two numbers. For example, the shaded region in Figure 2 corresponds to the probability that t is between a and b, written $P(a \le t \le b)$.

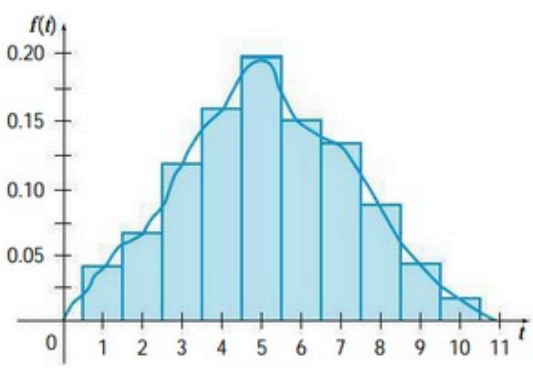


FIGURE 1

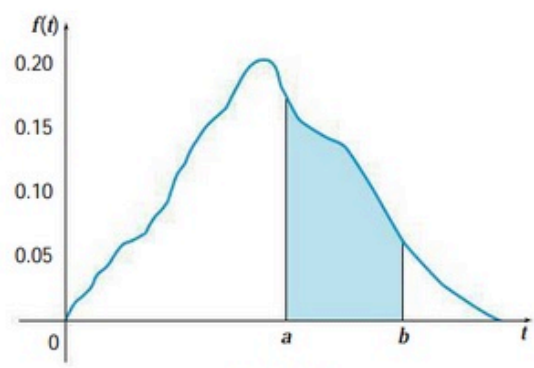


FIGURE 2

It was shown earlier that the definite integral of a continuous function f, where $f(x) \ge 0$, gives the area under the graph of $f(x)$ from $x = a$ to $x = b$. If a function f can be found to describe a continuous probability distribution, then the definite integral can be used to find the area under the curve from a to b that represents the probability that x will be between a and b.

If X is a continuous random variable whose distribution is described by the function f on $[a, b]$, then

$$P(a \le X \le b) = \int_a^b f(x)\, dx.$$

Probability Density Functions

A function f that describes a continuous probability distribution is called a *probability density function*. Such a function must satisfy the following conditions.

Probability Density Function

The function f is a **probability density function** of a random variable X in the interval $[a, b]$ if

1. $f(x) \geq 0$ for all x in the interval $[a, b]$; and
2. $\int_a^b f(x)\,dx = 1.$

Intuitively, Condition 1 says that the probability of a particular event can never be negative. Condition 2 says that the total probability for the interval must be 1; *something* must happen.

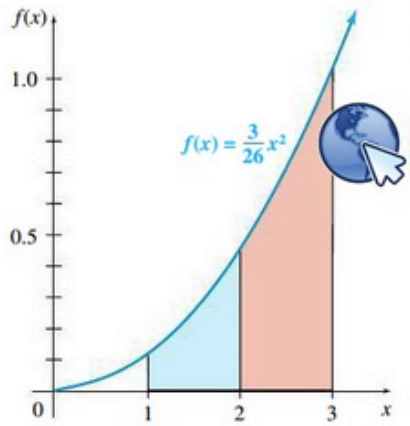


FIGURE 3

YOUR TURN 1 Repeat Example 1(a) for the function $f(x) = 2/x^2$ on $[1, 2]$. What is the probability that X will be between 3/2 and 2?

EXAMPLE 1 Probability Density Function

(a) Show that the function defined by $f(x) = (3/26)x^2$ is a probability density function for the interval $[1, 3]$.

SOLUTION First, note that Condition 1 holds; that is, $f(x) \geq 0$ for the interval $[1, 3]$. Next show that Condition 2 holds.

$$\int_1^3 \frac{3}{26}x^2 dx = \frac{x^3}{26}\bigg|_1^3 = \frac{27 - 1}{26} = 1$$

Since both conditions hold, $f(x)$ is a probability density function.

(b) Find the probability that X will be between 1 and 2.

SOLUTION The desired probability is given by the area under the graph of $f(x)$ between $x = 1$ and $x = 2$, as shown in blue in Figure 3. The area is found by using a definite integral.

$$P(1 \leq X \leq 2) = \int_1^2 \frac{3}{26}x^2\,dx = \frac{x^3}{26}\bigg|_1^2 = \frac{7}{26}$$

TRY YOUR TURN 1

Earlier, we noted that determining a suitable function is the most difficult part of applying mathematics to actual situations. Sometimes a function appears to model an application well but does not satisfy the requirements for a probability density function. In such cases, we may be able to change the function into a probability density function by multiplying it by a suitable constant, as shown in the next example.

EXAMPLE 2 Probability Density Function

Is there a constant k such that $f(x) = kx^2$ is a probability density function for the interval $[0, 4]$?

SOLUTION First,

$$\int_0^4 kx^2\,dx = \frac{kx^3}{3}\bigg|_0^4 = \frac{64k}{3}.$$

The integral must be equal to 1 for the function to be a probability density function. To convert it to one, let $k = 3/64$. The function defined by $(3/64)x^2$ for $[0, 4]$ will be a probability density function, since $(3/64)x^2 \geq 0$ for all x in $[0, 4]$ and

$$\int_0^4 \frac{3}{64}x^2 = 1.$$

TRY YOUR TURN 2

YOUR TURN 2 Repeat Example 2 for the function $f(x) = kx^3$ on the interval $[0, 4]$.

An important distinction is made between a discrete probability function and a probability density function (which is continuous). In a discrete distribution, the probability that the random variable, X, will assume a specific value is given in the distribution for every possible value of X. In a probability density function, however, the probability that X equals a specific value, say, c is

$$P(X = c) = \int_c^c f(x)\,dx = 0.$$

For a probability density function, only probabilities of *intervals* can be found. For example, suppose the random variable is the annual rainfall for a given region. The amount of rainfall in one year can take on any value within some continuous interval that depends on the region; however, the probability that the rainfall in a given year will be some specific amount, say 33.25 in., is actually zero.

The definition of a probability density function is extended to intervals such as $(-\infty, b]$, $(-\infty, b)$, $[a, \infty)$, (a, ∞), or $(-\infty, \infty)$ by using improper integrals, as follows.

Probability Density Functions on $(-\infty, \infty)$

If f is a probability density function for a continuous random variable X on $(-\infty, \infty)$, then

$$P(X \leq b) = P(X < b) = \int_{-\infty}^{b} f(x)\,dx,$$

$$P(X \geq a) = P(X > a) = \int_{a}^{\infty} f(x)\,dx,$$

$$P(-\infty < X < \infty) = \int_{-\infty}^{\infty} f(x)\,dx = 1.$$

The total area under the graph of a probability density function of this type must still equal 1.

FOR REVIEW

Improper integrals, those with one or two infinite limits, were discussed in Chapter 8 on Further Techniques and Applications of Integration. The type of improper integral we shall need was defined as

$$\int_a^\infty f(x)\,dx = \lim_{b\to\infty} \int_a^b f(x)\,dx.$$

For example,

$$\int_1^\infty x^{-2}\,dx = \lim_{b\to\infty} \int_1^b x^{-2}\,dx$$

$$= \lim_{b\to\infty} \left(-\frac{1}{x}\Big|_1^b\right)$$

$$= \lim_{b\to\infty} \left(-\frac{1}{b} + \frac{1}{1}\right)$$

$$= 0 + 1 = 1.$$

EXAMPLE 3 Location of a Bird's Nest

Suppose the random variable X is the distance (in kilometers) from a given point to the nearest bird's nest, with the probability density function of the distribution given by $f(x) = 2xe^{-x^2}$ for $x \geq 0$.

(a) Show that $f(x)$ is a probability density function.

SOLUTION Since $e^{-x^2} = 1/e^{x^2}$ is always positive, and $x \geq 0$,

$$f(x) = 2xe^{-x^2} \geq 0,$$

and Condition 1 holds.

Use substitution to evaluate the definite integral $\int_0^\infty 2xe^{-x^2}\,dx$. Let $u = -x^2$, so that $du = -2x\,dx$, and

$$\int 2xe^{-x^2}\,dx = -\int e^{-x^2}(-2x\,dx)$$
$$= -\int e^u\,du = -e^u = -e^{-x^2}.$$

Then

$$\int_0^\infty 2xe^{-x^2}\,dx = \lim_{b\to\infty}\int_0^b 2xe^{-x^2}\,dx = \lim_{b\to\infty}\left(-e^{-x^2}\right)\Big|_0^b$$
$$= \lim_{b\to\infty}\left(-\frac{1}{e^{b^2}} + e^0\right) = 0 + 1 = 1,$$

and Condition 2 holds.

The function defined by $f(x) = 2xe^{-x^2}$ satisfies the two conditions required of a probability density function.

(b) Find the probability that there is a bird's nest within 0.5 km of the given point.

APPLY IT

SOLUTION Find $P(X \le 0.5)$ where $X \ge 0$. This probability is given by

$$P(0 \le X \le 0.5) = \int_0^{0.5} 2xe^{-x^2}\,dx.$$

Now evaluate the integral. The indefinite integral was found in part (a).

$$P(0 \le X \le 0.5) = \int_0^{0.5} 2xe^{-x^2}\,dx = \left(-e^{-x^2}\right)\Big|_0^{0.5}$$
$$= -e^{-(0.5)^2} - (-e^0) = -e^{-0.25} + 1$$
$$\approx -0.7788 + 1 = 0.2212$$

The probability that a bird's nest will be found within 0.5 km of the given point is about 0.22.

TRY YOUR TURN 3

YOUR TURN 3 Using the probability density function of Example 3, find the probability that there is a bird's nest within 1 km of the given point.

Another important concept in probability is the *cumulative distribution function*, which gives the probability that a random variable X is less than or equal to an arbitrary value x.

Cumulative Distribution Function

If f is a probability density function of a random variable in the interval $[a, b]$, then the **cumulative distribution function** is defined as

$$F(x) = P(X \leq x) = \int_a^x f(t)\,dt$$

for $x \geq a$. Also, $F(x) = 0$ for $x < a$.

NOTE

1. We integrate with respect to the variable t in the integral, rather than x, because we are already using x for the upper limit on the integral. It doesn't matter what variable of integration is used in a definite integral, since that variable doesn't appear in the final answer. We just need to use a variable that's not being used for another purpose.
2. If the random variable is defined on the interval $(-\infty, \infty)$, simply replace a with $-\infty$ in the above definition.

EXAMPLE 5 Cumulative Distribution Function

Consider the random variable X defined in Example 3, giving the distance (in kilometers) from a given point to the nearest bird's nest, with probability density function $f(x) = 2xe^{-x^2}$ for $x \geq 0$.

(a) Find the cumulative distribution function for this random variable.

SOLUTION The cumulative distribution function is given by

$$F(x) = P(X \leq x) = \int_0^x 2te^{-t^2}dt \qquad \text{Use the density function with } t \text{ as the variable.}$$

$$= -e^{-t^2}\Big|_0^x \qquad \text{Use the antiderivative found in Example 3.}$$

$$= -e^{-x^2} + 1$$

for $x \geq 0$. The cumulative distribution function can be written as $F(x) = 1 - e^{-x^2}$ for $x \geq 0$. Note that for $x < 0$, $F(x) = 0$.

YOUR TURN 4 Use part (a) of Example 5 to calculate the probability that there is a bird's nest within 1 km of the given point.

(b) Use the solution to part (a) to calculate the probability that there is a bird's nest within 0.5 km of the given point.

SOLUTION To find $P(X \leq 0.5)$, calculate $F(0.5) = 1 - e^{-0.5^2} \approx 0.2212$. Notice that this is the same answer that we found in Example 3(b). TRY YOUR TURN 4

13.2 Expected Value and Variance of Continuous Random Variables

APPLY IT **What is the average age of a drunk driver in a fatal car crash?**
You will be asked to answer this question in Exercise 31.

It often is useful to have a single number, a typical or "average" number, that represents a random variable. The ***mean*** or *expected value* for a discrete random variable is found by multiplying each value of the random variable by its corresponding probability.

As we saw in the previous chapter, if x is a discrete random variable that takes on the n values, $x_1, x_2, \ldots, x_n$, with respective probabilities $p_1, p_2, \ldots, p_n$, then the mean, or expected value, of the random variable is

$$\mu = x_1p_1 + x_2p_2 + x_3p_3 + \cdots + x_np_n = \sum_{i=1}^{n} x_ip_i.$$

For the bird feeder example in the previous section, the expected value is given by

$$\begin{aligned}\mu &= 1(0.04) + 2(0.07) + 3(0.12) + 4(0.16) + 5(0.20) + 6(0.15) + 7(0.13)\\ &\quad + 8(0.08) + 9(0.04) + 10(0.01)\\ &= 5.09.\end{aligned}$$

Thus, the average time a sparrow spends at a bird feeder is 5.09 minutes.

This definition can be extended to continuous random variables by using definite integrals. Suppose a continuous random variable has probability density function f on $[a, b]$. We can divide the interval from a to b into n subintervals of length Δx, where $\Delta x = (b - a)/n$. In the ith subinterval, the probability that the random variable takes a value close to x_i is approximately $f(x_i)\Delta x$, and so

$$\mu \approx \sum_{i=1}^{n} x_i f(x_i)\Delta x.$$

As $n \to \infty$, the limit of this sum gives the expected value

$$\mu = \int_a^b xf(x)\,dx.$$

As we saw in the previous chapter, the *variance* of a discrete probability distribution is a measure of the *spread* of the values of the distribution. For a discrete distribution, the variance is found by taking the expected value of the squares of the differences of the values of the random variable and the mean. If the random variable x takes the values x_1, x_2, $x_3, \ldots, x_n$, with respective probabilities $p_1, p_2, p_3, \ldots, p_n$ and mean μ, then the variance of x is

$$\mathrm{Var}(x) = \sum_{i=1}^{n} (x_i - \mu)^2 p_i.$$

Think of the variance as the expected value of $(x - \mu)^2$, which measures how far x is from the mean μ. The *standard deviation* of x is defined as

$$\sigma = \sqrt{\mathrm{Var}(x)}.$$

For the bird feeder example in the previous section, the variance and standard deviation are

$$\begin{aligned}\mathrm{Var}(X) &= (1-5.09)^2(0.04) + (2-5.09)^2(0.07) + (3-5.09)^2(0.12)\\ &\quad + (4-5.09)^2(0.16) + (5-5.09)^2(0.20) + (6-5.09)^2(0.15)\\ &\quad + (7-5.09)^2(0.13) + (8-5.09)^2(0.08) + (9-5.09)^2(0.04)\\ &\quad + (10-5.09)^2(0.01)\\ &= 4.1819\end{aligned}$$

and,

$$\sigma = \sqrt{\mathrm{Var}(x)} \approx 2.045.$$

Like the mean or expected value, the variance of a continuous random variable is an integral.

$$\mathrm{Var}(x) = \int_a^b (x-\mu)^2 f(x)\,dx$$

To find the standard deviation of a continuous probability distribution, like that of a discrete distribution, we find the square root of the variance. The formulas for the expected value, variance, and standard deviation of a continuous probability distribution are summarized here.

Expected Value, Variance, and Standard Deviation

If x is a continuous random variable with probability density function f on $[a, b]$, then the **expected value** of x is

$$E(x) = \mu = \int_a^b xf(x)\,dx.$$

The **variance** of x is

$$\mathrm{Var}(x) = \int_a^b (x-\mu)^2 f(x)\,dx,$$

and the **standard deviation** of x is

$$\sigma = \sqrt{\mathrm{Var}(x)}.$$

NOTE In the formulas for expected value, variance, and standard deviation, and all other formulas in this section, it is possible that $a = -\infty$ or $b = \infty$, in which case the density function f is defined on $[a, \infty)$, $(-\infty, b]$, or $(-\infty, \infty)$. In this case, the integrals in these formulas become improper integrals, which are handled according to the procedure described in Section 8.4 on Improper Integrals. Example 2 will illustrate this procedure.

Geometrically, the expected value (or mean) of a probability distribution represents the balancing point of the distribution. If a fulcrum were placed at μ on the x-axis, the figure would be in balance. See Figure 6.

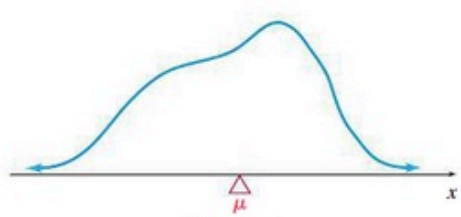


FIGURE 6

Calculating the variance in the last example was a messy job. An alternative version of the formula for the variance is easier to compute. This alternative formula is derived as follows.

$$\begin{aligned}\mathrm{Var}(x) &= \int_a^b (x-\mu)^2 f(x)\,dx \\ &= \int_a^b (x^2 - 2\mu x + \mu^2) f(x)\,dx \\ &= \int_a^b x^2 f(x)\,dx - 2\mu \int_a^b x f(x)\,dx + \mu^2 \int_a^b f(x)\,dx \qquad \textbf{(1)}\end{aligned}$$

By definition,

$$\int_a^b x f(x)\,dx = \mu,$$

and, since $f(x)$ is a probability density function,

$$\int_a^b f(x)\,dx = 1.$$

Substitute back into Equation (1) to get the alternative formula,

$$\mathrm{Var}(x) = \int_a^b x^2 f(x)\,dx - 2\mu^2 + \mu^2 = \int_a^b x^2 f(x)\,dx - \mu^2.$$

Alternative Formula for Variance

If x is a random variable with probability density function f on $[a, b]$, and if $E(x) = \mu$, then

$$\mathbf{Var}(x) = \int_a^b x^2 f(x)\,dx - \mu^2.$$

CAUTION Notice that the term μ^2 comes *after* the dx, and so is *not* integrated.

EXAMPLE 2 Variance

Use the alternative formula for variance to compute the variance of the random variable x with probability density function defined by $f(x) = 3/x^4$ for $x \geq 1$.

SOLUTION To find the variance, first find the expected value:

$$\begin{aligned}\mu &= \int_1^\infty x f(x)\,dx = \int_1^\infty x \cdot \frac{3}{x^4}\,dx = \int_1^\infty \frac{3}{x^3}\,dx \\ &= \lim_{b\to\infty} \int_1^b \frac{3}{x^3}\,dx = \lim_{b\to\infty} \left(\frac{3}{-2x^2}\right)\Bigg|_1^b = \frac{3}{2},\end{aligned}$$

or 1.5. Now find the variance by the alternative formula for variance:

$$\begin{aligned}
\mathrm{Var}(x) &= \int_1^\infty x^2\left(\frac{3}{x^4}\right)dx - \left(\frac{3}{2}\right)^2 \\
&= \int_1^\infty \frac{3}{x^2}\,dx - \frac{9}{4} \\
&= \lim_{b\to\infty}\int_1^b \frac{3}{x^2}\,dx - \frac{9}{4} \\
&= \lim_{b\to\infty}\left(\frac{-3}{x}\right)\Big|_1^b - \frac{9}{4} \\
&= 3 - \frac{9}{4} = \frac{3}{4}, \quad \text{or } 0.75.
\end{aligned}$$

YOUR TURN 2 Repeat Example 2 for the probability density function $f(x) = 4/x^5$ for $x \geq 1$.

TRY YOUR TURN 2

EXAMPLE 3 Patient Wait

A recent study has shown that the time patients wait at the doctor's office (in hours) after the time of their scheduled appointment is given by the probability density function $f(x) = 6t - 6t^2$ for $0 \leq t \leq 1$.

(a) Find and interpret the expected value for this distribution.

SOLUTION The expected value is

$$\begin{aligned}
\mu &= \int_0^1 t(6t - 6t^2)dt = \int_0^1 (6t^2 - 6t^3)dt \\
&= \left(2t^3 - \frac{3}{2}t^4\right)\Big|_0^1 = \frac{1}{2},
\end{aligned}$$

or 0.5. This result indicates that patients wait an average of 1/2 hour past the scheduled office appointment time.

(b) Compute the standard deviation.

SOLUTION First compute the variance. We use the alternative formula.

$$\begin{aligned}
\mathrm{Var}(t) &= \int_0^1 t^2(6t - 6t^2)dt - \left(\frac{1}{2}\right)^2 \\
&= \int_0^1 (6t^3 - 6t^4)dt - \left(\frac{1}{2}\right)^2 \\
&= \left(\frac{3}{2}t^4 - \frac{6}{5}t^5\right)\Big|_0^1 - \frac{1}{4} \\
&= \frac{3}{10} - \frac{1}{4} = \frac{1}{20} = 0.05
\end{aligned}$$

The standard deviation is $\sigma = \sqrt{0.05} \approx 0.22$.

(c) Calculate the probability that a patient's wait will be within one standard deviation of the mean.

SOLUTION Since the mean, or expected value, is 0.5 and the standard deviation is approximately 0.22, we are calculating the probability that a patient will wait between

$$\mu - \sigma = 0.5 - 0.22 = 0.28 \text{ hour}$$

and

$$\mu + \sigma = 0.5 + 0.22 = 0.72 \text{ hour.}$$

The probability is given by

$$P(0.28 \le T \le 0.22) = \int_{0.28}^{0.72} (6t - 6t^2)\, dt$$
$$= (3t^2 - 2t^3)\Big|_{0.28}^{0.72}$$
$$\approx 0.6174.$$

The probability that a patient's wait will be within one standard deviation of the mean is about 0.62.

TECHNO

EXAMPLE 4 Life Expectancy

In the previous section of this chapter we used statistics compiled by the National Center for Health Statistics to determine a probability density function that can be used to study the proportion of all 50-year-olds who will be alive in t years. The function is given by

$$S(t) = \frac{1}{497{,}703}(5.03958 \times 10^{-5}t^6 - 0.006603t^5 + 0.2992t^4 - 6.0507t^3 + 67.867t^2 - 110.3t + 2485.1)$$

for $0 \le t \le 52.5$.

(a) Find the life expectancy of a 50-year-old person.

SOLUTION Since this is a complicated function that is tedious to integrate analytically, we will employ the integration feature on a TI-84 Plus calculator to calculate

$$\mu = \int_0^{52.5} tS(t)dt \approx 30.38 \text{ years.}$$

According to life tables, the life expectancy of a person between the ages of 50 and 55 is 30.6 years. Our estimate is remarkably accurate given the limited number of data points and the function used in our original analysis. Life expectancy is generally calculated with techniques from life table analysis. ***Source: National Center for Health Statistics.***

(b) Find the standard deviation of this probability function.

SOLUTION Using the alternate formula, we first calculate the variance.

$$\text{Var}(T) = \int_0^{52.5} t^2S(t)dt - \mu^2 = 1057.7195 - (30.38)^2 \approx 134.775$$

Thus, $\sigma = \sqrt{\text{Var}(T)} \approx 11.61$ years.

As we mentioned earlier, the expected value is also referred to as the mean of the random variable. It is a type of average. There is another type of average, known as the *median*, that is often used. It is the value of the random variable for which there is a 50% probability of being larger and a 50% probability of being smaller. The precise definition is as follows.

Median

If X is a random variable with probability density function f on $[a, b]$, then the **median** of X is the number m such that

$$\int_a^m f(x)\,dx = \frac{1}{2}.$$

The median is particularly useful when the random variable is not distributed symmetrically about the mean. An example of this would be a random variable representing the price of homes in a city. There is a small probability that a home will be much more expensive than most of the homes in the city, and this tends to make the mean abnormally high. The median price is a better representation of the average price of a home.

EXAMPLE 5 Median

Find the median for the random variable described in Example 2, with density function defined by $f(x) = 3/x^4$ for $x \geq 1$.

SOLUTION According to the formula,

$$\int_1^m \frac{3}{x^4}\,dx = \frac{1}{2}.$$

Evaluating the integral on the left, we have

$$-\frac{1}{x^3}\bigg|_1^m = -\frac{1}{m^3} + 1.$$

Set this equal to 1/2.

$$\begin{aligned} -\frac{1}{m^3} + 1 &= \frac{1}{2} \\ \frac{1}{2} &= \frac{1}{m^3} && \text{Subtract } 1/2 \text{ from both sides, and add } 1/m^3. \\ m^3 &= 2 && \text{Cross multiply.} \\ m &= \sqrt[3]{2} \end{aligned}$$

The median value is, therefore, $\sqrt[3]{2} \approx 1.2599$. Notice that this is smaller than the mean of 1.5 found in Example 2. This is because the random variable can take on arbitrarily large values, which increases the mean but doesn't affect the median. TRY YOUR TURN 3

YOUR TURN 3 Repeat Example 5 for the probability density function $f(x) = 4/x^5$ for $x \geq 1$.

Using the notion of cumulative distribution function from the previous section, we can say that the median m is the value for which the cumulative distribution function is 0.5; that is, $F(m) = 0.5$.

13.2 EXERCISES

In Exercises 1–8, a probability density function of a random variable is defined. Find the expected value, the variance, and the standard deviation. Round answers to the nearest hundredth.

1. $f(x) = \frac{1}{4}$; $[3, 7]$
2. $f(x) = \frac{1}{10}$; $[0, 10]$
3. $f(x) = \frac{x}{8} - \frac{1}{4}$; $[2, 6]$
4. $f(x) = 2(1 - x)$; $[0, 1]$
5. $f(x) = 1 - \frac{1}{\sqrt{x}}$; $[1, 4]$
6. $f(x) = \frac{1}{11}\left(1 + \frac{3}{\sqrt{x}}\right)$; $[4, 9]$
7. $f(x) = 4x^{-5}$; $[1, \infty)$
8. $f(x) = 3x^{-4}$; $[1, \infty)$
9. What information does the mean (expected value) of a continuous random variable give?
10. Suppose two random variables have standard deviations of 0.10 and 0.23, respectively. What does this tell you about their distributions?

38. Social Network In Exercise 46 of the previous section, the probability density function for the number of U.S. users of Facebook, an online social network, was found to be

$$S(t) = \frac{1}{466.26}(-0.00007445t^4 + 0.01243t^3 - 0.7419t^2 + 18.18t - 137.5)$$

where t was the number of years since birth on $[13.4, 62.0]$. Calculate the expected age of a Facebook user, as well as the standard deviation. ***Source: Inside Facebook.***

39. Earthquakes The time between major earthquakes in the Southern California region is a random variable with probability density function defined by

$$f(t) = \frac{1}{960}e^{-t/960},$$

where t is measured in days. ***Source: Journal of Seismology.*** Find the expected value and the standard deviation of this probability density function.

40. Annual Rainfall The annual rainfall in a remote Middle Eastern country is a random variable with probability density function defined by

$$f(x) = \frac{5.5 - x}{15}, \quad \text{for } x \text{ in } [0, 5].$$

a. Find the mean annual rainfall.

b. Find the standard deviation.

c. Find the probability of a year with rainfall less than 1 standard deviation below the mean.

YOUR TURN ANSWERS

1. 4/3, 0.0706
2. 2/9
3. $\sqrt[4]{2} \approx 1.1892$

13.3 Special Probability Density Functions

APPLY IT

What is the probability that the maximum outdoor temperature will be higher than 24°C? What is the probability that a flashlight battery will last longer than 40 hours?

These questions, presented in Examples 1 and 2, can be answered if the probability density function for the maximum temperature and for the life of the battery are known.

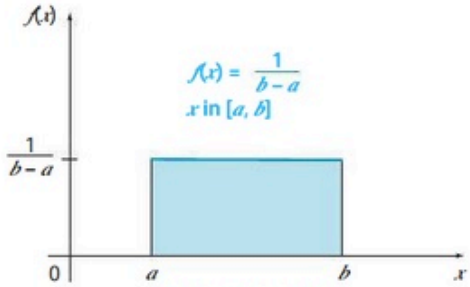
In practice, it is not feasible to construct a probability density function for every experiment. Instead, a researcher uses one of several probability density functions that are well known, matching the shape of the experimental distribution to one of the known distributions. In this section we discuss some of the most commonly used probability distributions.

Uniform Distribution

The simplest probability distribution occurs when the probability density function of a random variable remains constant over the sample space. In this case, the random variable is said to be *uniformly distributed* over the sample space. The probability density function for the **uniform distribution** is defined by

$$f(x) = \frac{1}{b - a} \quad \text{for } x \text{ in } [a, b],$$

where a and b are constant real numbers. The graph of $f(x)$ is shown in Figure 9.



Uniform distribution

FIGURE 9

Since $b - a$ is positive, $f(x) \geq 0$, and

$$\int_a^b \frac{1}{b-a}\,dx = \frac{1}{b-a}x\Big|_a^b = \frac{1}{b-a}(b-a) = 1.$$

Therefore, the function is a probability density function.

The expected value for the uniform distribution is

$$\mu = \int_a^b \left(\frac{1}{b-a}\right)x\,dx = \left(\frac{1}{b-a}\right)\frac{x^2}{2}\Big|_a^b$$

$$= \frac{1}{2(b-a)}(b^2 - a^2) = \frac{1}{2}(b+a). \qquad b^2 - a^2 = (b-a)(b+a)$$

The variance is given by

$$\begin{aligned}
\mathrm{Var}(X) &= \int_a^b \left(\frac{1}{b-a}\right)x^2\,dx - \left(\frac{b+a}{2}\right)^2 \\
&= \left(\frac{1}{b-a}\right)\frac{x^3}{3}\Big|_a^b - \frac{(b+a)^2}{4} \\
&= \frac{1}{3(b-a)}(b^3 - a^3) - \frac{1}{4}(b+a)^2 \\
&= \frac{b^2 + ab + a^2}{3} - \frac{b^2 + 2ab + a^2}{4} && b^3 - a^3 = (b-a)(b^2 + ab + a^2) \\
&= \frac{b^2 - 2ab + a^2}{12}. && \text{Get a common denominator; subtract.}
\end{aligned}$$

Thus

$$\mathrm{Var}(X) = \frac{1}{12}(b-a)^2, \qquad \text{Factor.}$$

and

$$\sigma = \frac{1}{\sqrt{12}}(b-a).$$

These properties of the uniform distribution are summarized below.

Uniform Distribution

If X is a random variable with probability density function

$$f(x) = \frac{1}{b-a} \quad \text{for } x \text{ in } [a, b],$$

then

$$\mu = \frac{1}{2}(b+a) \quad \text{and} \quad \sigma = \frac{1}{\sqrt{12}}(b-a).$$

EXAMPLE 1 Daily Temperature

A couple is planning to vacation in San Francisco. They have been told that the maximum daily temperature during the time they plan to be there ranges from 15°C to 27°C. Assume that the probability of any temperature between 15°C and 27°C is equally likely for any given day during the specified time period.

(a) What is the probability that the maximum temperature on the day they arrive will be higher than 24°C?

APPLY IT

SOLUTION If the random variable T represents the maximum temperature on a given day, then the uniform probability density function for T is defined by $f(t) = 1/12$ for the interval $[15, 27]$. By definition,

$$P(T > 24) = \int_{24}^{27} \frac{1}{12}\,dt = \frac{1}{12}t\Big|_{24}^{27} = \frac{1}{4}.$$

(b) What average maximum temperature can they expect?

SOLUTION The expected maximum temperature is

$$\mu = \frac{1}{2}(27 + 15) = 21,$$

or 21°C.

(c) What is the probability that the maximum temperature on a given day will be one standard deviation or more below the mean?

SOLUTION First find σ.

$$\sigma = \frac{1}{\sqrt{12}}(27 - 15) = \frac{12}{\sqrt{12}} = \sqrt{12} = 2\sqrt{3} \approx 3.464.$$

One standard deviation below the mean indicates a temperature of $21 - 3.464 = 17.536$°C.

$$P(T \le 17.536) = \int_{15}^{17.536} \frac{1}{12}\,dt = \frac{1}{12}t\Big|_{15}^{17.536} \approx 0.2113$$

The probability is about 0.21 that the temperature will not exceed 17.5°C.

TRY YOUR TURN 1

YOUR TURN 1 The next vacation for the couple in Example 1 is to a desert with a maximum daily temperature that uniformly ranges from 27°C to 42°C. Find the expected maximum temperature and the probability that the maximum temperature will be within one standard deviation of the mean.

Exponential Distribution

The next distribution is very important in reliability and survival analysis. When manufactured items and living things have a constant failure rate over a period of time, the exponential distribution is used to describe their probability of failure. In this case, the random variable is said to be *exponentially distributed* over the sample space. The probability density function for the **exponential distribution** is defined by

$$f(x) = ae^{-ax} \qquad \text{for } x \text{ in } [0, \infty),$$

where a is a positive constant. The graph of $f(x)$ is shown in Figure 10.

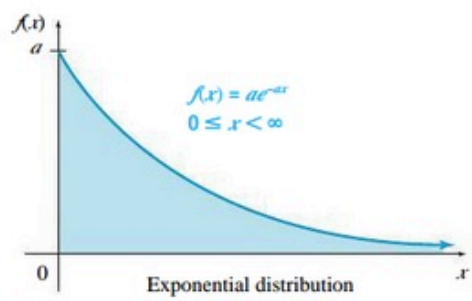


FIGURE 10

Here $f(x) \ge 0$, since e^{-ax} and a are both positive for all values of x. Also,

$$\int_0^\infty ae^{-ax}\,dx = \lim_{b\to\infty}\int_0^b ae^{-ax}\,dx$$

$$= \lim_{b\to\infty}(-e^{-ax})\Big|_0^b = \lim_{b\to\infty}\left(\frac{-1}{e^{ab}} + \frac{1}{e^0}\right) = 1,$$

so the function is a probability density function.

The expected value and the standard deviation of the exponential distribution can be found using integration by parts. The results are given below. (See Exercise 28 at the end of this section.)

Exponential Distribution

If X is a random variable with probability density function

$$f(x) = ae^{-ax} \quad \text{for } x \text{ in } [0, \infty),$$

then

$$\mu = \frac{1}{a} \quad \text{and} \quad \sigma = \frac{1}{a}.$$

EXAMPLE 2 Flashlight Battery

Suppose the useful life (in hours) of a flashlight battery is the random variable T, with probability density function given by the exponential distribution

$$f(t) = \frac{1}{20}e^{-t/20} \quad \text{for } t \geq 0.$$

(a) Find the probability that a particular battery, selected at random, has a useful life of less than 100 hours.

SOLUTION The probability is given by

$$\begin{aligned} P(T \leq 100) &= \int_0^{100} \frac{1}{20}e^{-t/20}\,dt = \frac{1}{20}(-20e^{-t/20})\Big|_0^{100} \\ &= -(e^{-100/20} - e^0) = -(e^{-5} - 1) \\ &\approx 1 - 0.0067 = 0.9933. \end{aligned}$$

(b) Find the expected value and standard deviation of the distribution.

SOLUTION Use the formulas given above. Both μ and σ equal $1/a$, and since $a = 1/20$ here,

$$\mu = 20 \quad \text{and} \quad \sigma = 20.$$

This means that the average life of a battery is 20 hours, and no battery lasts less than 1 standard deviation below the mean.

(c) What is the probability that a battery will last longer than 40 hours?

SOLUTION The probability is given by

$$P(T > 40) = \int_{40}^{\infty} \frac{1}{20}e^{-t/20}\,dt = \lim_{b\to\infty}(-e^{-t/20})\Big|_{40}^{b} = \frac{1}{e^2} \approx 0.1353,$$

or about 14%.

TRY YOUR TURN 2

APPLY IT

YOUR TURN 2 Repeat Example 2 for a flashlight battery with a useful life given by the probability density function $f(t) = \frac{1}{25}e^{-t/25}$ for $t \geq 0$.

There is an interesting connection between the exponential random variable and the discrete random variable known as the *Poisson distribution*, which is used to describe events that occur randomly at a constant rate, such that the probability of two occurrences in a very short time interval is essentially 0. The Poisson distribution is used to describe DNA mutations (per unit length of a strand of DNA, rather than unit time), radioactive decay, defects that occur sporadically at random (as opposed to defects that occur because of something becoming old and worn out), and many other phenomena.

Poisson Distribution

If X is a random variable with the **Poisson distribution**, then the probability that X has the value x is given by

$$P(X = x) = f(x) = \frac{a^x e^{-a}}{x!},$$

where $a > 0$ and $x = 0, 1, 2, \ldots$. For the Poisson distribution,

$$\mu = a \quad \text{and} \quad \sigma = \sqrt{a}.$$

EXAMPLE 3 Chromosomal Aberrations

Researchers have used the Poisson distribution to model the number of chromosomal aberrations in cultures of human leukocytes treated with a chemical mutagen. ***Source: SIAM Journal on Applied Mathematics.*** (The researchers found that a more complicated model based on the Poisson distribution gave a better fit. Interested readers should see the original article as listed in the Sources.) They estimated that $a = 0.5696$ aberration per unit time. Find the probability of the following number of aberrations per unit time.

(a) At most 1

SOLUTION According to the Poisson distribution, with $a = 0.5696$,

$$\begin{aligned} P(\text{at most } 1) &= P(0) + P(1) \\ &= \frac{0.5696^0 e^{-0.5696}}{0!} + \frac{0.5696^1 e^{-0.5696}}{1!} \\ &= 0.56575 + 0.32225 \\ &= 0.8880. \end{aligned}$$

There is about a 89% chance of at most 1 aberration.

(b) More than 1

SOLUTION

$$\begin{aligned} P(\text{more than } 1) &= 1 - P(\text{at most } 1) \\ &= 1 - 0.8880 \\ &= 0.1120 \end{aligned}$$

The connection between the Poisson distribution, which is discrete, and the exponential distribution, which is continuous, is that the time until the next occurrence of a Poisson random variable with parameter a is an exponential random variable with parameter a.

EXAMPLE 4 Chromosomal Aberrations

For the cultures discussed in Example 3, what is the probability that at least 2 time periods pass before the next aberration?

SOLUTION Because the number of aberrations has a Poisson distribution with $a = 0.5696$, the waiting time until the next aberration is distributed exponentially with $a = 0.5696$.

$$\begin{aligned} P(X > 2) &= \int_2^\infty 0.5696 e^{-0.5696t}\, dt \\ &= \lim_{b\to\infty} \int_2^b 0.5696 e^{-0.5696t}\, dt \\ &= \lim_{b\to\infty} \left(-e^{-0.5696t} \Big|_2^b \right) \\ &= \lim_{b\to\infty} \left(\frac{-1}{e^{0.5696b}} + \frac{1}{e^{0.5696(2)}} \right) \\ &\approx 0.3201 \end{aligned}$$

Normal Distribution

The **normal distribution**, with its well-known bell-shaped graph, is undoubtedly the most important probability density function. It is widely used in various applications of statistics. The random variables associated with these applications are said to be normally distributed. The probability density function for the normal distribution has the following characteristics.

Normal Distribution

If μ and σ are real numbers, $\sigma > 0$, and if X is a random variable with probability density function defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)} \quad \text{for } x \text{ in } (-\infty, \infty),$$

then

$$E(X) = \mu \quad \text{and} \quad \text{Var}(X) = \sigma^2, \quad \text{with standard deviation } \sigma.$$

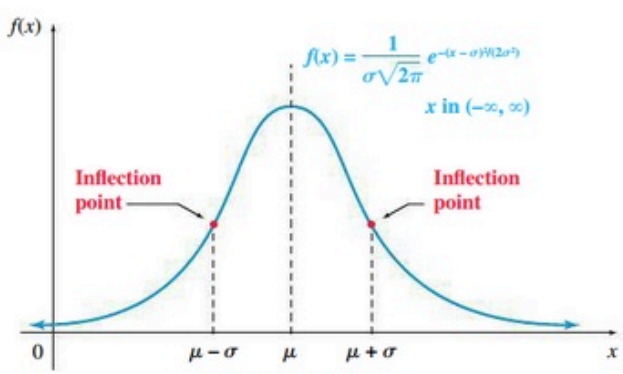
Notice that the definition of the probability density function includes σ, which is the standard deviation of the distribution.

Advanced techniques can be used to show that

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}\, dx = 1.$$

Deriving the expected value and standard deviation for the normal distribution also requires techniques beyond the scope of this text.

Each normal probability distribution has associated with it a bell-shaped curve, called a **normal curve**, such as the one in Figure 11. Each normal curve is symmetric about a vertical line through the mean, μ. Vertical lines at points $+1\sigma$ and -1σ from the mean show the inflection points of the graph. (See Exercise 30 at the end of this section.) A normal curve never touches the x-axis; it extends indefinitely in both directions.



Normal distribution

FIGURE 11

The development of the normal curve is credited to the Frenchman Abraham De Moivre (1667–1754). Three of his publications dealt with probability and associated topics: *Annuities upon Lives* (which contributed to the development of actuarial studies), *Doctrine of Chances*, and *Miscellanea Analytica*.

Many different normal curves have the same mean. In such cases, a larger value of σ produces a "flatter" normal curve, while smaller values of σ produce more values near the mean, resulting in a "taller" normal curve. See Figure 12 on the next page.